

Lab 6: Managing File System Permissions in RHEL 9

Section 1: List and Interpret File System Permissions

Question: List the permissions for the file /home/student/project.txt. What do the permission bits mean for the owner, group, and others?

```
-rw-rw-r-- 1 Student Student 0 Dec 17 07:14 project.txt
```

Answer : users [can read and write] , group [read and write] , other [read only]

Question: List the permissions for the directory /home/student/. What effect do the permissions have on access to this directory by the owner, group, and others?

```
drwxr-x--- 2 Student Student 4096 Dec 17 07:14 .
```

Answer : users [read , write , execute] , group[read and execute] , others[can do nothing]

Bonus Question: Explain the difference between read, write, and execute permissions on files versus directories.

Answer :

1 – Files :

- **Read** → less , cat , more
- **Write** → vim , nano ,
- **execute** → executable file , can be execute

2 – Directories:

- **Read** → ls
- **Write** → ls , touch , mkdir
- **execute** → cd

Section 2: Change Permissions and Ownership of Files

Question1 : Use the chmod command to give the user full permissions (read, write, execute), the group read and execute permissions, and others no permissions on the file /home/student/project.txt.

Answer :

```
Student@abdo:~$ chmod 750 project.txt
Student@abdo:~$ ls -al project.txt
-rwxr-x--- 1 Student Student 0 Dec 17 07:14 project.txt
Student@abdo:~$
```

Question2 :

Change the owner of the file /home/student/project.txt to the user admin and the group to developers.

Answer :

- **chown Admin:Developers project.txt**

Bonus Question: Set the permission of /home/student/project.txt to rwxr-x--- using the symbolic mode in the chmod command.

Answer :

- **chmod u+rwx,g+rx project.txt**

Section 3: Default File Permissions and Special Permissions

Question 1 :

Create a new directory /home/student/shared. Set the default permission using umask so that any file created in this directory by any user has no permissions for others.

```
Student@abdo:~/Shared$ umask 006
Student@abdo:~/Shared$ touch Test_File.txt
Student@abdo:~/Shared$ ls -al
total 8
drwxr-xr-x 2 Student Student 4096 Dec 17 07:47 .
drwxr-x--- 3 Student Student 4096 Dec 17 07:45 ..
-rw-rw---- 1 Student Student   0 Dec 17 07:47 Test_File.txt
Student@abdo:~/Shared$
```

Question: Add the sticky bit to the /home/student/shared directory to ensure that users can only delete their own files within this directory.

Answer :

chmod 1755 Shared/

Bonus Question: Explain what happens when a directory has the sticky bit enabled. How does it affect file deletion?

Answer :

When sticky bit enabled it prevent any user has access on Directory to delete any thing in the directory , stikybit applied on the Directories only

Section 4: Set Group Ownership of Files Using Special Permissions

Question: Create a new directory /home/student/groupdir. Set the setgid (Set Group ID) permission on this directory, so that files created in this directory inherit the group ownership of the directory.

Answer : chmod 2771 groupdir/

Question: Verify that when a new file is created in /home/student/groupdir, it has the correct group ownership based on the directory's group ownership.

Bonus Question: Explain how the setgid permission affects group ownership of files and directories.

Answer : The program runs with the group ID of the file owner, not the group of the user who executes i

Section 5: Control Default Permissions of New Files

Question: Set the default umask for the user student to 027. Verify that any new file created by this user has the correct permissions.

Question: Create a new file and verify that the default permissions reflect the new umask settings.

```
Student@abdo:~/GroupDir$ cd ..
Student@abdo:~$ umask 027
Student@abdo:~$ touch hello.txt
Student@abdo:~$ ls -ld hello.txt
-rw-r----- 1 Student Student 0 Dec 17 08:40 hello.txt
Student@abdo:~$
```

Bonus Question: What is the default umask value in most Linux systems, and how does it affect the creation of files and directories?

The default value of the most linux system is 0022

when happen on file → the current permission number is subtracted from **666**

when happen on Dir → the current permission number is subtracted from **777**